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Math Coaching Assistant



Solve for x : $x^2 - 7x + 12 = 0$



To solve the equation $x^2 - 7x + 12 = 0$, we can try to factor the quadratic expression. What is the first step in finding the factors?



Subtract 12 from both sides



While you can subtract 12 from both sides, it's usually easier to solve a quadratic equation when it's set to zero. To factor $x^2 - 7x + 12 = 0$, we need two numbers that multiply to 12 and add to -7. Which pair of numbers satisfies both conditions?



3 and 4



The numbers 3 and 4 add up to 7, but the equation requires them to add up to -7. How should we adjust the signs of these numbers so their product remains 12 but their sum becomes -7?



-3 and -4



That's correct, since $(-3) \cdot (-4) = 12$ and $(-3) + (-4) = -7$. The equation in factored form is $(x - 3)(x - 4)$

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 That's correct, since $(-3) \cdot (-4) = 12$ and $(-3) + (-4) = -7$. The equation in factored form is $(x - 3)(x - 4) = 0$. Based on the Zero Product Property, what must be true for the product to equal zero?

 $x + 3 = 0$ or $x + 4 = 0$

 That is incorrect. The factors we found were $(x - 3)$ and $(x - 4)$, so those are the terms that must be set to zero. Which of the following correctly applies the Zero Product Property to $(x - 3)(x - 4) = 0$?

 $x = -3$ or $x = -4$

 That is incorrect. If you substitute $x = -3$ into $(x - 3)$, the result is -6 , not 0 . The current equation is $(x - 3)(x - 4) = 0$. What value of x makes the first factor, $(x - 3)$, equal to zero?

 $x = -3$

 That is incorrect. If we replace x with -3 in the expression $x - 3$, we get $-3 - 3 = -6$, which is not zero. For the equation $x - 3 = 0$ to be true, what value must x have?

 -3

 If we have the equation $x - 3 = 0$, we want to find a value that makes the statement true. To isolate x , what operation should we perform on both sides of the equation?

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Subtract 3 from both sides



Subtracting 3 from both sides is incorrect because it results in $x - 6 = -3$, which does not isolate x . To solve the equation $x - 3 = 0$, we need to perform the inverse operation of subtraction. What should we do to both sides to isolate x ?



Add 3 to both sides



That's correct. Adding 3 to both sides of $x - 3 = 0$ gives $x = 3$. Now we need to solve the second part of the equation: $x - 4 = 0$. What is the value of x in this case?

 $x = -4$ 

That is incorrect. If we substitute -4 for x in the expression $x - 4$, we get $-4 - 4 = -8$, which is not zero. To solve the equation $x - 4 = 0$, what operation should we perform on both sides to isolate x ?



Subtract 4 from both sides



That is incorrect. Subtracting 4 from both sides of $x - 4 = 0$ would result in $x - 8 = -4$, which does not isolate x . To see why, what is the result of the calculation $-4 - 4$?

 -8

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 $x = -4$



That is incorrect. If we substitute -4 for x in the expression $x - 4$, we get $-4 - 4 = -8$, which is not zero. To solve the equation $x - 4 = 0$, what operation should we perform on both sides to isolate x ?



Subtract 4 from both sides



That is incorrect. Subtracting 4 from both sides of $x - 4 = 0$ would result in $x - 8 = -4$, which does not isolate x . To see why, what is the result of the calculation $-4 - 4$?



-8



Correct, $-4 - 4$ is -8 , so subtracting 4 did not isolate x . To get x by itself in the equation $x - 4 = 0$, we need to undo the subtraction of 4.



Add 4 to both sides



That's correct; adding 4 to both sides gives $x = 4$. We have now found both solutions: $x = 3$ and $x = 4$. To solve this, we first factored the quadratic into $(x - 3)(x - 4) = 0$ by finding numbers that multiply to 12 and add to -7 . Then, we applied the Zero Product Property to set each factor to zero and used inverse operations (adding 3 and 4) to isolate x . Remember, to undo subtraction in an equation like $x - 4 = 0$, you must perform the inverse operation, which is addition.

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